Overview

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8.3 Calibration

8.1 Information, groups, and domains

- We consider **integration of information** about the population with the aim of:
  - stratification or unequal-probability sampling ($\Rightarrow$ efficiency gains),
  - adjusting for frame imperfections and nonresponse.
- Information is available in the form of **unit-level auxiliary variables** and known **population totals** or **group counts**.
- We distinguish the **stage** at which the information is incorporated (role):
  - **Design stage** (sampler), e.g., information from the sampling frame for identifying domains/strata
  - **Estimation stage** (analyst) $\Rightarrow$ post-stratification/calibration
- Information is often used at **both** stages.

Example 20 (Ophthalmologists).

Ophthalmologists by language region and gender (MedReg, 2022)

Language region	Count
German	1'260
French	572
Italian	70

Language region	Gender	Count
German	Female	416
	Male	844
French	Female	189
	Male	383
Italian	Female	26
	Male	44

Definition 14 (Grouping).

A **grouping** $\mathscr{G}$ of the population U is a partition

$$\mathscr{G} = \{U_1, \ldots, U_G\}, \qquad U = \bigcup_{g=1}^{G} U_g, \qquad U_g \cap U_{g'} = \varnothing \quad (g \neq g').$$

- We use **grouping** as a generic term. Depending on when the groups are defined, they may represent design strata, post-strata, or domains.
- Let $\mathscr{H}$ denote the stratification used at the design stage. We distinguish, for example:
  - $\mathscr{G} = \mathscr{H}$: the groups coincide with the design strata;
  - $\mathscr{G}$ **refines** $\mathscr{H}$: each group U_g is contained in one design stratum.

8.2 Post-stratification

- Let $\mathscr{G} = \{U_1, \ldots, U_G\}$ denote a grouping of U with known group sizes N_g.
- The **post-stratified estimator** of the population mean $\bar{y}_U$ is

$$\bar{y}^{\text{post}} = \sum_{g=1}^{G} W_g \, \bar{y}_g, \qquad W_g = \frac{N_g}{N}, \qquad \bar{y}_g = \frac{1}{n_g} \sum_{i \in s_g} y_i.$$

- If $\mathscr{G} = \mathscr{H}$, the post-strata coincide with the design strata and $\bar{y}^{\text{post}} = \bar{y}^{\text{strat}}$.
- Otherwise, the post-stratum sample sizes

$$n_g(S) = |S \cap U_g|$$

are **random variables**.

- In many applications, $\mathscr{G}$ refines the design stratification $\mathscr{H}$.
- Suppose the original sample is an **SRS** of fixed size n.
  - The post-stratum sample sizes

    $$n_g(S) = |S \cap U_g|$$

    are random, with

    $$\mathbb{E}_p\{n_g(S)\} = nW_g.$$

  - It is possible that $n_g(S) = 0$ for some post-stratum. This becomes unlikely when the expected post-stratum sample sizes nW_g are sufficiently large.
  - In practice, post-strata with no or very few sampled units may need to be **combined**.

- If the post-stratum sample sizes are sufficiently large, an **approximate variance estimator** is (Särndal et al., 1992, Remark 7.10.6)

$$\widehat{\mathrm{var}}_p(\bar{y}^{\mathrm{post}}) \approx \underbrace{(1-f)\sum_{g=1}^{G} W_g \frac{\widehat{\sigma}_g^2}{n}}_{=\widehat{\mathrm{var}}_p^{\mathrm{prop}}(\bar{y}^{\mathrm{post}})} + \underbrace{\frac{1-f}{n}\sum_{g=1}^{G}(1-W_g)\frac{\widehat{\sigma}_g^2}{n}}_{\text{additional contribution from random post-stratum sample sizes}} .$$

  $\Rightarrow$ The approximation is understood **conditional** on $n_g(S) > 0$ for all $g = 1, \ldots, G$.

- The second term is of smaller order and becomes **negligible** as the post-stratum sample sizes increase. Hence,

$$\widehat{\mathrm{var}}_p(\bar{y}^{\mathrm{post}}) \approx \widehat{\mathrm{var}}_{p_{\mathrm{prop}}}(\bar{y}^{\mathrm{post}}).$$

- For sufficiently large post-stratum sample sizes, an SRS with post-stratification is approximately as **efficient** as stratified SRS with **proportional allocation** (Holt and Smith, 1979).

- Stratification at the design stage is useful when we want to
  - improve **efficiency**, or
  - control the sample sizes for important **domains**.

- Post-stratification is more **flexible**: different groupings can be used at the estimation stage for different variables or purposes.

- This flexibility is particularly useful in **multipurpose surveys**.

8.3 Calibration

- Let $w_i = 1/\pi_i$ denote the **design weight** of sampled unit $i = 1, \ldots, n$.
- **Calibration** replaces the design weights w_i by **calibration weights** w_i^* that (Deville and Särndal, 1992)
  1. remain as close as possible to the design weights, and
  2. satisfy the **calibration constraints**

$$\sum_{i \in s} w_i^* \mathbf{x}_i = \mathbf{t}_x, \qquad \mathbf{t}_x = \sum_{i \in U} \mathbf{x}_i,$$

  where $\mathbf{x}_i = (x_{i1}, \ldots, x_{iq})^T$ is a vector of auxiliary variables.

- Thus, the weighted sample reproduces the known **population totals** of the auxiliary variables.

- The **calibration estimator** of the population total t_y is

$$\widehat{t}_y^{cal} = \sum_{i \in s} w_i^* y_i.$$

- Regarding the available **auxiliary information**, we distinguish two cases:

  Case 1) $\mathbf{x}_i$ is known for all $i \in U$; hence, the population total $\mathbf{t}_x$ is also known.

  Case 2) $\mathbf{x}_i$ is known only for $i \in s$, but the population total $\mathbf{t}_x$ is known from an external source.

- For calibration, the **essential requirement** is that $\mathbf{x}_i$ is available for the sampled units and that $\mathbf{t}_x$ is known.

Examples of choices for $\mathbf{x}_i$ (and the corresponding population totals)

Case	Auxiliary vector ($i \in s$)	Known population total
1	1	N
2	x_i	t_x
3	$(1, x_i)^T$	$(N, t_x)^T$
4	$\left(\mathbb{1}_i^D, \mathbb{1}_i^F, \mathbb{1}_i^I\right)^T$	$\left(N_D, N_F, N_I\right)^T$

with indicator variables for language region: $\mathbb{1}_i^D = 1$ if the unit is German-speaking and 0 otherwise; analogously for the French (F) and Italian (I) language regions. N_D, N_F, and N_I denote the corresponding population sizes.

Example 21 (Language Regions).

Calibration with

$$\mathbf{x}_i = \left(1, \mathbb{1}_i^D, \mathbb{1}_i^F, \mathbb{1}_i^I\right)^T \qquad \text{and} \qquad \mathbf{t}_x = \left(N, N_D, N_F, N_I\right)^T$$

leads to **linearly dependent calibration variables**.

- Let $\mathbf{X}$ denote the $(n \times q)$ **matrix** given by $\mathbf{X} = (\mathbf{x}_1^T, \ldots, \mathbf{x}_n^T)^T$. In this example, $\mathbf{X}$ does **not have full column rank**.
- We can therefore either
  - omit the constant 1 from $\mathbf{x}_i$ and N from $\mathbf{t}_x$, or
  - omit one language-region indicator and the corresponding population count.

- The calibration constraints generally do not determine the weights w_i^* uniquely. We therefore choose weights that remain **as close as possible** to the design weights w_i.

- Deville and Särndal (1992) consider several distance functions. One choice is the **chi-square distance**

$$D_{\chi^2}(\mathbf{w}, \mathbf{w}^*) = \sum_{i \in s} \frac{(w_i^* - w_i)^2}{q_i w_i}, \qquad q_i > 0.$$

- The calibration weights are obtained by solving

$$\min_{\mathbf{w}^*} D_{\chi^2}(\mathbf{w}, \mathbf{w}^*) \qquad \text{subject to} \qquad \sum_{i \in s} w_i^* \mathbf{x}_i = \mathbf{t}_x.$$

- In what follows, we set $q_i \equiv 1$.

- For the chi-square distance, the **calibration weights** have the closed-form solution

$$w_i^* = w_i \left[1 + \left(\mathbf{t}_x - \widehat{\mathbf{t}}_x^{\mathrm{HT}} \right)^T \left(\sum_{j \in s} w_j \mathbf{x}_j \mathbf{x}_j^T \right)^{-1} \mathbf{x}_i \right],$$

where

$$\widehat{\mathbf{t}}_x^{\mathrm{HT}} = \sum_{i \in s} w_i \mathbf{x}_i$$

is the Horvitz–Thompson estimator of $\mathbf{t}_x$.

- The resulting **calibration estimator** of t_y can be written as

$$\widehat{t}_y^{cal} = \widehat{t}_y^{\mathrm{HT}} + \left(\mathbf{t}_x - \widehat{\mathbf{t}}_x \right)^T \widehat{\mathbf{b}}, \qquad \text{where} \qquad \widehat{\mathbf{b}} = \left(\sum_{i \in s} w_i \mathbf{x}_i \mathbf{x}_i^T \right)^{-1} \sum_{i \in s} w_i \mathbf{x}_i y_i.$$

- The calibration estimator $\widehat{t}_y^{cal}$
  - is generally **not design-unbiased**, but is design consistent under suitable regularity conditions;
  - can be more **efficient** than the HT estimator when the auxiliary variables are informative about y;
  - coincides, for the chi-square distance, with the **generalized regression estimator** (GREG).
- Deville and Särndal (1992) showed that calibration estimators based on a broad class of **distance functions** are asymptotically equivalent.
- Variance estimation can be based on the residuals $y_i - \mathbf{x}_i^T \widehat{\mathbf{b}}$.

Why use calibration?

- Calibration provides a systematic way to incorporate **auxiliary information** into estimation.
- Important purposes include:
  - improving **efficiency** when the auxiliary variables are informative about y;
  - ensuring **consistency with known population totals**;
  - reducing the effects of **nonresponse** or frame imperfections.
- For nonresponse adjustment, useful auxiliary variables should be related to the **response propensity** and/or the **variables of interest**; see Särndal and Lundström (2005).

Historical context

- The basic idea of adjusting survey weights to incorporate known population information predates Deville and Särndal (1992).
- An important early example is **iterative proportional fitting** (raking), introduced by Deming and Stephan (1940) to adjust contingency tables to known marginal totals.
- Deville and Särndal (1992) provided a general framework for **calibration weighting**, based on calibration constraints and distance functions.
- **Entropy balancing** (Hainmueller, 2012), widely used in the social sciences, can be viewed as a special case of calibration weighting using an entropy-type distance function.

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ETH zürich

My contact details at ETH
Lecturer at the Department of Mathematics
E-mail: tobias.schoch@math.ethz.ch

Primary affiliation
University of Applied Sciences Northwestern Switzerland FHNW
School of Business – Institute ICC
Riggenbachstrasse 16
CH-4600 Olten

Prof. Dr. Tobias Schoch
E-mail: tobias.schoch@fhnw.ch
Phone (direct): +41 (0)62 957 21 02

Version of October 21, 2026